

PYTHON BEGINNER CHEAT SHEET

Core Fundamentals • Variables • Input & Output • Decisions • Loops

1. Variables & Types

Basics

Variables store information in memory using a descriptive name.

```
# Text (String)
name = "Mariam"

# Whole Number (Integer)
age = 15

# Decimal (Float)
score = 95.5

# True/False (Boolean)
is_active = True
```

2. Output: print()

Display

Displays text or variable values on the screen.

```
# Basic print
print("Hello World!")

# f-Strings (Embed variables directly)
print(f"Agent {name} is {age} years old.")

# Print with custom separator & ending
print("2026", "08", "09", sep="-")
print("Loading", end="...")
```

3. Input: input()

Interactive

Pauses code to get user input. Always returns a **String**.

```
# Get text input
city = input("Where do you live? ")

# Convert input to Integer for math
count = int(input("Enter a number: "))

# Convert input to Float for decimals
price = float(input("Enter price: "))
```

Rule: Wrap input() in int() or float() to do math!

4. Conditionals

Decisions

Runs specific code based on conditions (True/False).

```
score = 85

if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
else:
    print("Keep Practicing!")

# Operators: ==, !=, >, <, >=, <=
# Combine with: and, or, not
```

5. For Loops

Counting

Repeats code a set number of times or through a collection.

```
# Repeat 3 times (0, 1, 2)
for i in range(3):
    print(f"Iteration {i}")

# Loop through a list
for fruit in ["apple", "banana"]:
    print(fruit)
```

6. While Loops

Until True

Keeps running as long as a condition stays True.

```
# Loop until user enters secret password
while True:
    guess = input("Enter secret code: ")
    if guess == "1234":
        print("Unlocked!")
        break # Exits the loop
```